

COUNTING FINITE GROUPS

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1. Introduction

Let $G(n)$ denote the number of non-isomorphic groups of order n . We can ask whether there is a formula for $G(n)$ in terms of the arithmetic structure of n . In the absence of such a formula, we can ask if there are upper and lower bounds for $G(n)$. We can study the growth of $G(n)$. Interestingly enough, these questions have partial (and, in some cases, complete) answers. Their investigations take us into the domain of both finite group theory and analytic number theory. It is the purpose of this article to survey these investigations and to adumbrate problems and topics for further research.

At the outset, we have the trivial estimate

$$(1) \quad G(n) \leq n^{n^2}$$

simply by counting all of the possible multiplication tables. This can be substantially improved. With the classification of the finite simple groups, one knows from the work of Neumann [N] that if $\omega(n)$ denotes the total number of prime factors of n , counted with multiplicity, then

$$(2) \quad G(n) \leq n^{\omega(n)^2}$$

This is a substantial improvement because by a famous result of Hardy and Ramanujan [HR], $\omega(n)$ has normal order $\log \log n$. Even otherwise, $\omega(n)$ has a much smaller order than n . It is trivial to see that $\omega(n) \leq (\log n)/\log 2$, and Ramanujan [R] has shown that

$$(3) \quad \omega(n) = O\left(\frac{\log n}{\log \log n}\right).$$

This gives the estimate

$$(4) \quad \log G(n) = O\left(\frac{\log^3 n}{(\log \log n)^2}\right).$$

Recently, McIver and Neumann [MN] obtained the estimate:

$$G(n) \leq n^{\omega(n)^2/2}$$

This result has been improved by Pyber [Py] in the following way. Let $p_1, \dots, p_k$ be distinct primes and let $\mathcal{P} = \{P_1, \dots, P_k\}$ be a set of groups such that P_i is a p_i -group of order $p_i^{g_i}$ (say). We say that G is a group with Sylow set $\mathcal{P}$ if

$$|G| = \prod_{i=1}^k p_i^{g_i}$$

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and the Sylow p_i -subgroups of G are isomorphic to P_i . Pyber [Py] has proved that the number of groups of order

$$n = \prod_{i=1}^k p_i^{g_i}$$

with a given Sylow set $\mathcal{P}$ is at most

$$n^{75\mu+16}$$

where

$$\mu = \max g_i.$$

As a corollary, he obtains

$$G(n) \leq n^{(\frac{2}{27} + o(1))\mu^2}$$

as

$$\mu \rightarrow \infty.$$

However, these estimates do not reflect the truth because usually $G(n)$ is much smaller. For example, when $\mu = 1$, we shall see below that $G(n) \leq n$. If we confine our attention to various classes of groups, like p -groups, or if we constrain the arithmetic nature of n , like squarefree numbers or numbers with bounded prime power exponents, then these estimates are sharp.

2. Counting abelian groups.

Since every finite abelian group is isomorphic to a direct product of cyclic groups of prime power order, it is easy to derive an explicit formula for $G_A(n)$, denoting the number of abelian groups of order n . If $p(\alpha)$ denotes the number of partitions of α , then

$$(5) \quad G_A(n) = p(\alpha_1)p(\alpha_2) \dots p(\alpha_r)$$

where

$$(6) \quad n = p_1^{\alpha_1} \dots p_r^{\alpha_r}$$

is the unique factorization of n into distinct prime powers. We see that $G_A(n)$ is a multiplicative function of n . As such, the Dirichlet series attached to $G_A(n)$ has a beautiful expression in terms of the classical Riemann ζ -function:

$$(7) \quad \sum_{n=1}^{\infty} \frac{G_A(n)}{n^s} = \prod_{k=1}^{\infty} \zeta(ks)$$

Therefore, the methods of analytic number theory can be utilised to study

$$\sum_{n \leq x} G_A(n)$$

In fact, noting that the right hand side of (7) is analytic for $\text{Re}(s) \geq 1$ except for a simple pole at $s = 1$, we get by the Wiener-Ikehara Tauberian theorem (see for example [M, p. 37]) that

$$(8) \quad \sum_{n \leq x} G_A(n) \sim C_1 x$$

where

$$C_1 = \prod_{k=2}^{\infty} \zeta(k).$$

Kendall and Rankin [KR] proved

$$(9) \quad \sum_{n \leq x} G_A(n) = C_1 x + C_2 x^{1/2} + O(x^{1/3} \log^2 x)$$

with

$$C_2 = \prod_{k=3}^{\infty} \zeta(k/2).$$

One can improve these results further by invoking subtler properties of the Riemann ζ -function and we will not go into the details here. We will refer the reader to the paper by Balasubramanian and Ramachandra [BR] where the oscillation of the error term is investigated.

3. Counting p -groups. If p is prime, there are reasonable estimates for $G(p^n)$ due to Higman and Sims. For instance, Sims [S] proves that

$$\log G(p^n) = n^3 \left(\frac{2}{27} + O(n^{-1/3}) \right)$$

as $n \rightarrow \infty$ and with p fixed.

Let $G_N(n)$ denote the number of nilpotent groups of order n (up to isomorphism). Since every nilpotent group is isomorphic to a direct product of its Sylow subgroups, we find that

$$G_N(n) = \prod_{p^\alpha \parallel n} G(p^\alpha)$$

where the notation $p^\alpha \parallel n$ means that $p^\alpha \mid n$ and $p^{\alpha+1} \nmid n$. This multiplicative property however is unwieldy to give any asymptotic information on the sum

$$\sum_{n \leq x} G_N(n).$$

At present, we have no such asymptotic formula. Neither do we have a proof of the plausible conjecture:

$$\lim_{x \rightarrow \infty} \frac{\sum_{n \leq x} G_N(n)}{\sum_{n \leq x} G(n)} = 0,$$

which is the assertion that the set of nilpotent groups has ‘density’ zero in the set of all finite groups.

4. Groups of squarefree order.

When n is squarefree, one can give a remarkable explicit formula for $G(n)$. Apparently, this formula was originally discovered by O. Hölder [H] in 1895. It was rediscovered by the authors in [MM] almost a century later.

Let $v(p, n)$ denote the number of prime divisors q of n which are $\equiv 1 \pmod{p}$. Then, for n squarefree,

$$(10) \quad G(n) = \sum_{d|n} \prod_{p|d} \frac{p^{v(p, n/d)} - 1}{p - 1}.$$

The authors in [MM] derive a more general formula. Define $v(p^j, m)$ by

$$p^{v(p^j, m)} = \prod_{q|m} (p^j, q - 1)$$

where p and q denote prime numbers. If m is squarefree, this coincides with the definition of $v(p, m)$ above.

Let us call a C-group a finite group all of whose Sylow subgroups are cyclic. Thus any group of squarefree order is a C-group. Let $C(n)$ denote the number of non-isomorphic C-groups of order n . Then the formula

$$(11) \quad C(n) = \sum_{\substack{d|n \\ (d, n/d)=1}} \prod_{p^\alpha || d} \left(\sum_{j=1}^{\alpha} \frac{p^{v(p^j, n/d)} - p^{v(p^{j-1}, n/d)}}{p^{j-1}(p - 1)} \right)$$

is derived in [MM]. Letting $g(n)$ be defined by

$$(12) \quad g(n) = \prod_{p|n} (n, p - 1),$$

one can show that

$$(13) \quad C(n) \leq g(n).$$

In particular, if n is squarefree, we deduce

$$(14) \quad G(n) \leq \phi(n)$$

where ϕ denotes Euler’s ϕ -function.

This is a substantial improvement over the trivial bound (1) and the general non-trivial bound (2). By means of these equalities and inequalities, one can derive the asymptotic behaviour of the sums

$$\sum_{n \leq x} \log C(n)$$

and

$$\sum_{n \leq x} \mu^2(n) \log G(n).$$

More precisely, let

$$\wedge(n) = \begin{cases} \log p & \text{if } n = p^a \\ 0 & \text{otherwise} \end{cases}$$

be the familiar von Mangoldt function of number theory. Then the asymptotic formulas

$$(15) \quad \sum_{n \leq x} \log C(n) = \left(\sum_{n=1}^{\infty} \frac{\wedge(n)}{n\phi(n)} + o(1) \right) x \log \log x$$

and

$$(16) \quad \sum_{n \leq x} \mu^2(n) \log G(n) = \left(\frac{6}{\pi^2} \sum_p \frac{\log p}{p^2 - 1} + o(1) \right) x \log \log x$$

as $x \rightarrow \infty$ are proved in [MM].

In [MM], the authors asked if the upper bound (14) is sharp. If not, we must answer the question: if n ranges over squarefree numbers, is it true that

$$(17) \quad G(n) = o(\phi(n))?$$

More generally, is it true that

$$(18) \quad C(n) = o(\phi(n))$$

as n tends to infinity? This conjecture was proved by R. Murty and S. Srinivasan [MS] in 1987. They established that for some constant $A > 0$,

$$(19) \quad C(n) = O\left(\frac{\phi(n)}{(\log n)^A \log \log \log n}\right).$$

Moreover, this is the true order (apart from the constant A) of $C(n)$. The authors in [MS] prove that there is a constant $B > 0$ so that

$$(20) \quad G(n) > \frac{\phi(n)}{(\log n)^B \log \log \log n}$$

for infinitely many squarefree numbers n .

In this context, it is to be observed that from the explicit formulae (10) and (11), the asymptotic behavior of

$$\sum_{n \leq x} C(n)$$

and

$$\sum_{n \leq x} \mu^2(n) G(n)$$

seems difficult to determine. The result (19) allows us to deduce

$$(21) \quad \sum_{n \leq x} C(n) = o(x^2)$$

as $x \rightarrow \infty$. Pomerance [P] has shown that the asymptotic behaviour of the sum in (21) is intimately connected with the Halberstam-Elliott conjecture about the distribution of prime numbers. If $\pi(x, q, a)$ denotes the number of primes $p \leq x$ such that $p \equiv a \pmod{q}$, then the Halberstam-Elliott conjecture predicts that for any $\epsilon > 0$,

$$(22) \quad \sum_{q < x^{1-\epsilon}} \left| \pi(x, q, a) - \frac{\text{li } x}{\phi(q)} \right| \ll \frac{x}{\log^A x}$$

for any $A > 0$. Let us recall the definition of $\text{li } x$:

$$\text{li } x = \int_2^x \frac{dt}{\log t}.$$

The Halberstam-Elliott conjecture is to be viewed as an extension of the celebrated theorem of Bombieri-Vinogradov which asserts that an estimate of the form (21) holds if we sum over $q < x^{1/2-\epsilon}$.

Thus, assuming (22), Pomerance [P] shows that

$$(22) \quad \sum_{n \leq x} C(n) > x^{2-\epsilon}.$$

Utilizing a key theorem of Balog-Fouvry-Rousselet, Pomerance shows unconditionally that

$$(23) \quad \sum_{n \leq x} \mu^2(n) G(n) \geq x^{1.68}.$$

He conjectures that

$$(24) \quad \sum_{n \leq x} \mu^2(n) G(n) = x^2 \exp\left(-\left(1 + o(1)\right) \frac{(\log x)(\log \log \log x)}{\log \log x}\right).$$

The upper bound implied by (24) has been established in [P].

5. Value distribution of $G(n)$.

Let $F_k(x)$ denote the number of $n \leq x$ such that $G(n) = k$. For $k = 1$, this counts the number of $n \leq x$ such that all groups of order n are cyclic. By group theory, one can show that any group of order n is cyclic if and only if $(n, \phi(n)) = 1$.

Using the sieve of Eratosthenes and Brun's sieve, Erdős [E] showed that the number of such $n \leq x$ satisfies as $x \rightarrow \infty$

$$(25) \quad F_1(x) \sim \frac{x e^{-\gamma}}{\log \log \log x},$$

where γ is Euler's constant. The authors in [MM2] proved that

$$(26) \quad F_2(x) = O\left(\frac{x \log \log \log \log x}{(\log \log \log x)^2}\right)$$

and conjectured that

$$(27) \quad F_2(x) \sim \frac{x e^{-\gamma}}{(\log_3 x)^2}$$

where $\log_k x = \log(\log_{k-1} x)$ and $\log_1 x = \log x$.

This conjecture was subsequently proved by Kumar Murty, Ram Murty and P. Erdős in [EMM]. They also observed a striking disparity in the asymptotic behaviour of $F_k(x)$ when k is a power of 2 and when k is not. If $k = 2^a$, they proved

$$(28) \quad F_k(x) = \frac{e^{-\gamma}(1 + o(1))x}{a! (\log \log \log x)^{a+1}}$$

and if $k \neq 2^a$, then

$$(29) \quad F_k(x) \ll \frac{x}{(\log \log x)^{1-\epsilon}}.$$

They suggested that with some care (29) can be improved to

$$F_k(x) = o\left(\frac{x}{\log \log x}\right).$$

Thus, $G(n)$ prefers to take powers of 2 more often than not.

In the same paper, they remark that for n squarefree,

$$\frac{\log G(n)}{\log \log n}$$

has a continuous distribution function.

6. Distribution of integers attached to various classes of groups.

Let $\mathcal{D}$ be a class of finite groups like

- $\mathcal{C}$: cyclic groups
- $\mathcal{A}$: abelian groups
- $\mathcal{N}$: nilpotent groups
- $\mathcal{SS}$: supersolvable groups
- $\mathcal{S}$: solvable groups

Perhaps it will be convenient for the reader if we briefly recall the definitions of these concepts. A finite group G is said to be nilpotent if the upper central series terminates at G . That is, we define inductively (using the correspondence theorem):

$$Z_0(G) = 1, \quad Z_i(G)/Z_{i-1}(G) = Z(G/Z_{i-1}(G)) \quad i \geq 1$$

where $Z(G)$ denotes the center of G and say that G is nilpotent if for some i , $Z_i(G) = G$. A finite group is called supersolvable if there is a chain of subgroups

$$1 = G_0 \subseteq G_1 \subseteq G_2 \subseteq \cdots \subseteq G_m = G$$

such that each G_i is normal in G and G_{i+1}/G_i is cyclic of prime order. A finite group is called solvable if in the above, each G_i is required to be normal in G_{i+1} . Thus, supersolvability is a stronger condition than solvability.

Let $H_{\mathcal{D}}(x)$ denote the number of $n \leq x$ such that any group of order n belongs to $\mathcal{D}$. For example, by the result of Erdős [E] cited in the previous section,

$$H_{\mathcal{C}}(x) = \frac{e^{-\gamma}(1 + o(1))x}{\log_3 x}$$

as $x \rightarrow \infty$. In [MM3], the authors consider $H_{\mathcal{A}}(x)$, $H_{\mathcal{N}}(x)$, $H_{\mathcal{SS}}(x)$, $H_{\mathcal{S}}(x)$ and prove that

$$(30) \quad H_{\mathcal{N}}(x) = \frac{e^{-\gamma}(1 + o(1))x}{\log_3 x}$$

as $x \rightarrow \infty$. Thus $H_{\mathcal{C}}(x)$, $H_{\mathcal{A}}(x)$ and $H_{\mathcal{N}}(x)$ have the same asymptotic behaviour as $x \rightarrow \infty$.

In the case of solvable and supersolvable groups, they establish the existence of certain positive constants C_1 and C_2 so that

$$(31) \quad H_{\mathcal{SS}}(X) \sim C_1 x$$

and

$$(32) \quad H_S(x) \sim C_2 x$$

as $x \rightarrow \infty$.

7. Concluding remarks.

Can one derive the asymptotic behaviour of

$$\sum_{n \leq x} G_s(n)$$

where $G_s(n)$ denotes the number of solvable groups of order n ? Is it true that

$$\lim_{x \rightarrow \infty} \frac{\sum_{n \leq x} G_s(n)}{\sum_{n \leq x} G(n)} = 0,$$

implying that the set of finite solvable groups has 'density' zero?

These questions often involve group theory to reduce the problem to one amenable by sieve theory. Perhaps the methods indicated in this paper as well as those listed in the references will be useful in answering them. In any event, these investigations reveal the exciting interplay between finite group theory and analytic number theory.

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